Matrix Representation of Graph

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Graphs are useful tools to show how things are connected, like cities linked by roads or people connected in a social network. A graph has vertices (points) and edges (lines connecting the points).

One simple way to work with graphs is by using matrices. A matrix is a table of numbers that can show which vertices are connected. This is called the matrix representation of a graph. It helps to store graph information clearly and makes it easier to use in computer programs or calculations.

There are different types of matrix representations, like the adjacency matrix and the incidence matrix, each useful for different types of problems. Using matrices makes it easier to understand and solve questions about graphs in a simple and organized way.

1 Incidence Matrix of a Connected Graph

Let G be a graph with n vertices and e edges. Suppose G contains no self loops. We define a matrix $A = (a_{ij})_{n\times e}$, whose rows correspond to vertices and columns correspond to the edges, as follows:

$$a_{ij} = \begin{cases} 1, & \text{if the } j\text{-th edge } e_j \text{ is incident on the } i\text{-th vertex } v_i \\ 0, & \text{otherwise.} \end{cases}$$

This matrix A is called the **Incidence Matrix** of G and denoted as $A(G)$.

Note 1.1. *If the graph G contains no parallel edge then this matrix is known as* ***adjacency matrix****.*

Example 1.2. *Find the incidence matrix of the following graph.*

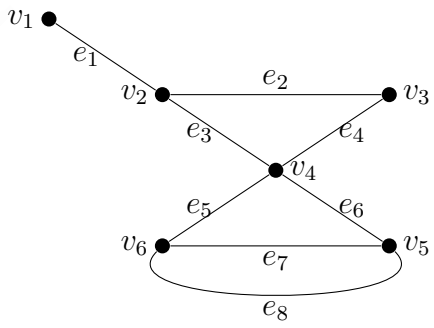


Fig. 1.1

Solution. This graph has six vertices and eight edges, therefore the incidence matrix is of order 6×8.

$$\mathbf{A(G)} = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{cccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

Example 1.3. *Draw a graph whose incidence matrix is*

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Note 1.4. *(i) All the entries of an incidence matrix are either 0 or 1.*

(ii) The number of 1's in each row is equal to the degree of the corresponding vertex.

(iii) A row with all zeros corresponds to an isolated vertex.

(iv) Each column of the incidence matrix has exactly two 1's, since every edge is incident on exactly two vertices.

(v) Identical columns in the incidence matrix correspond to parallel edges.

(vi) Swapping rows or columns, change the order of the vertices or order of the edges, but does not change the actual structure of the graph.

Example 1.5. *Does there exist a graph corresponding to the following incidence matrix? Justify.*

$$A(G) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution. No, because the second column has no 1 (each column should contain exactly two 1s except for self-loop).

2 Incidence Matrix of a connected Digraph

Let G be a Digraph with n vertices, e edges. Suppose G contains no self-loops. We define a matrix $A = (a_{ij})_{n \times e}$ whose rows correspond to the vertices and whose columns correspond to the edges, as follows:

$$a_{ij} = \begin{cases} 1, & \text{if the } j\text{-th edge is incident } \textbf{out} \text{ of the } i\text{-th vertex} \\ -1, & \text{if the } j\text{-th edge is incident } \textbf{into} \text{ the } i\text{-th vertex} \\ 0, & \text{if the } j\text{-th edge is neither incident out nor incident into the } i\text{-th vertex} \end{cases}$$

This matrix A is called the **Incidence Matrix** of the digraph $G$and is denoted by $A(G)$.

Example 2.1. *Write the incidence matrix of the following digraph:*

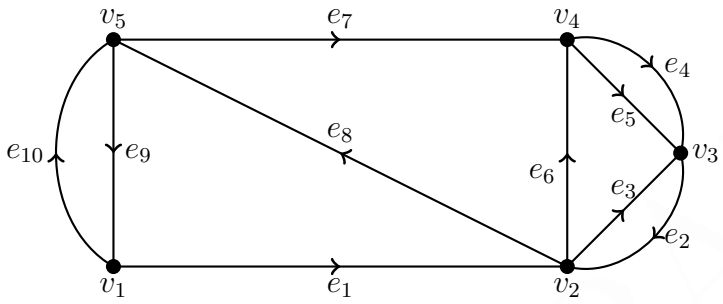


Fig. 8.25

Solution. There are 5 vertices and 10 edges in the graph. So the incident matrix will be of 5×10 order. Here v_1 and v_2 are the initial and terminal vertices of e_1, respectively. v_2 and v_3 are the initial and terminal vertices of e_2, respectively, and so on. Therefore, the required incidence matrix is:

$$\mathbf{A(G)} = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{cccccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 & e_9 & e_{10} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 & 1 & 1 & 1 & -1 \end{bmatrix} \end{array}$$

Example 2.2. *Draw a graph whose incidence matrix is*

$$\begin{bmatrix} 0 & 0 & 1 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

Note 2.3. *The incident matrix f the corresponding undirected graph is obtained by changing all* -1 *entries to* 1.

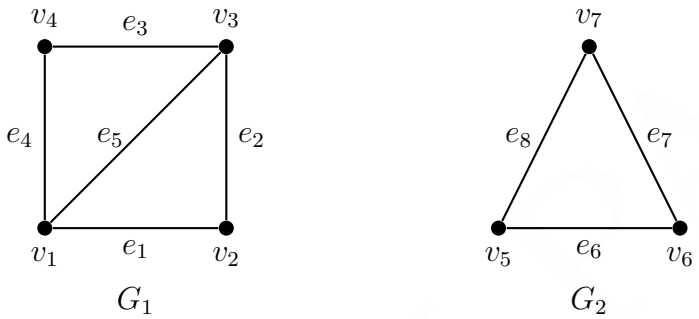
3 Incidence Matrix of Disconnected Graph/Digraph

Let G be a disconnected graph/digraph having two components, G_1 and G_2. Then the Incidence Matrix of G can be written in a Block-Diagonal form as

$$A(G) = \begin{bmatrix} A(G_1) & O \\ O & A(G_2) \end{bmatrix}$$

where $A(G_1)$, $A(G_2)$ are incidence matrices of G_1 and G_2 respectively and O is null matrix.

Example 3.1. *Obtain the incident matrix of the following matrix G having two components G_1 and G_2:*



Solution. The incident matrix of the component G_1 is

$$\mathbf{A(G_1)} = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

The incident matrix of the component G_2 is $\mathbf{A(G_2)} = \begin{array}{c} \\ v_5 \\ v_6 \\ v_7 \end{array} \begin{array}{c} \begin{array}{ccc} e_6 & e_7 & e_8 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{array}$

Therefore, the incident matrix of the given graph G is

$$\mathbf{A(G)} = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \\ v_7 \end{array} \begin{array}{c} \begin{array}{cccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$